Kernix 0.11.1 / Performance brief

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

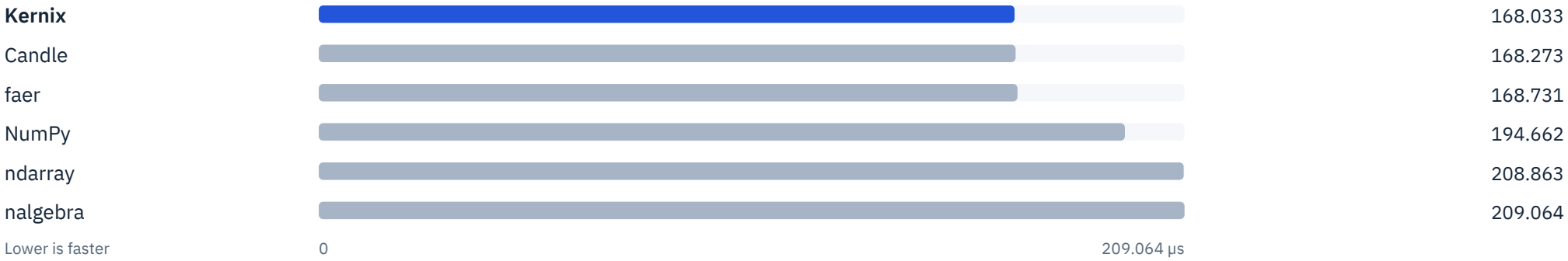
35 workloads

119 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.11.1 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



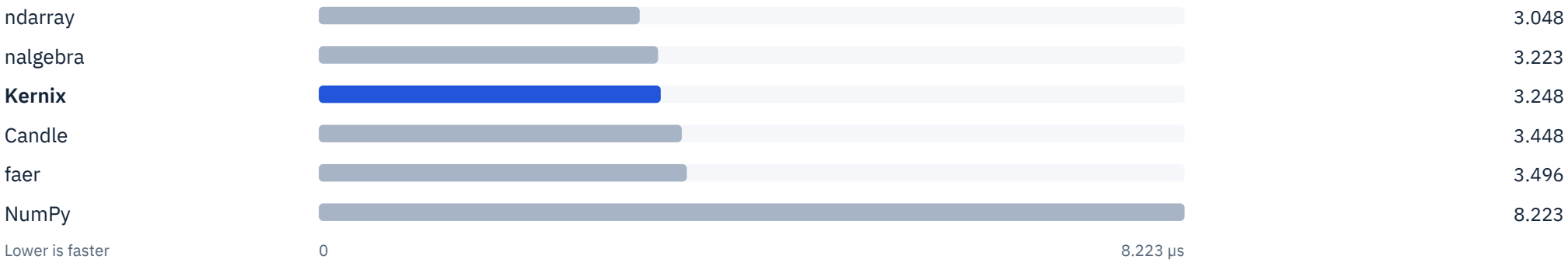
Kernix 0.11.1 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



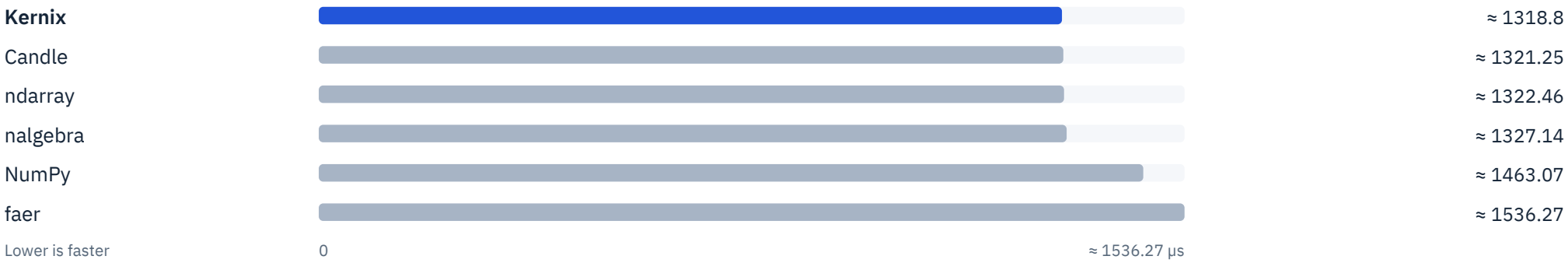
Kernix 0.11.1 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

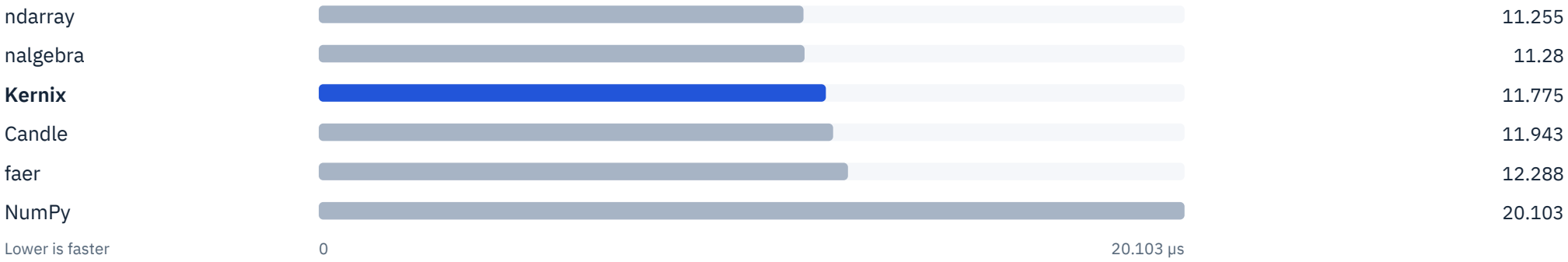
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



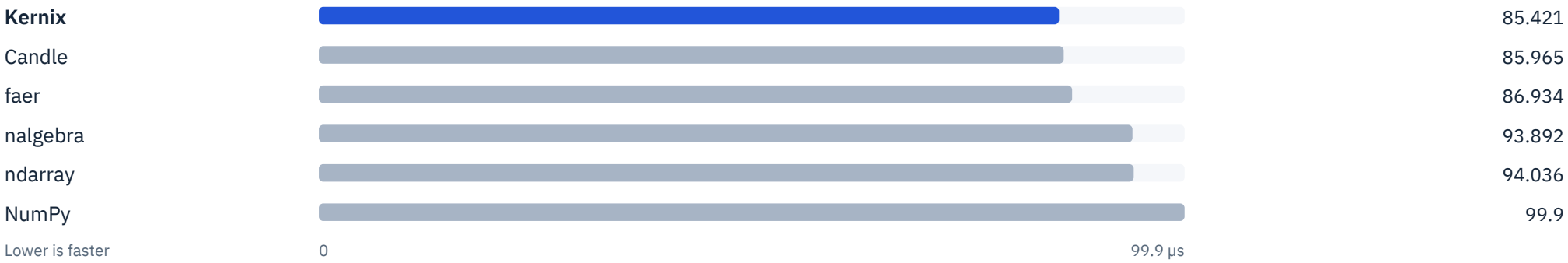
Kernix 0.11.1 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

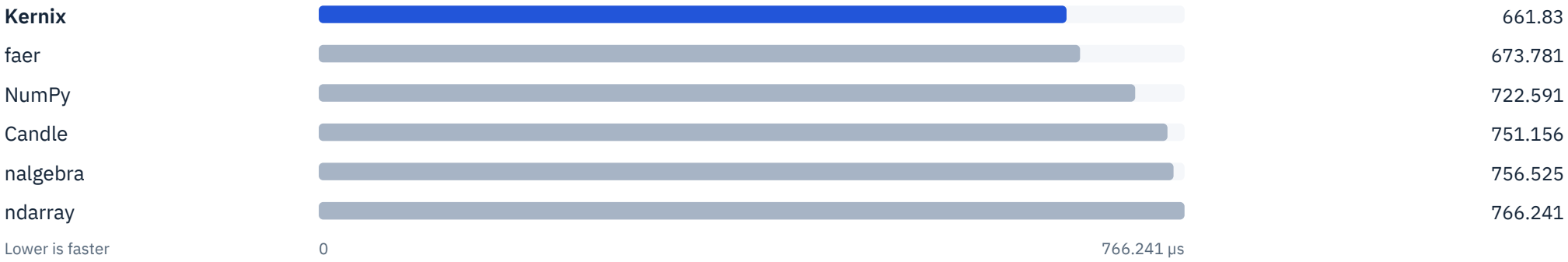
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



Kernix 0.11.1 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

Per-call median

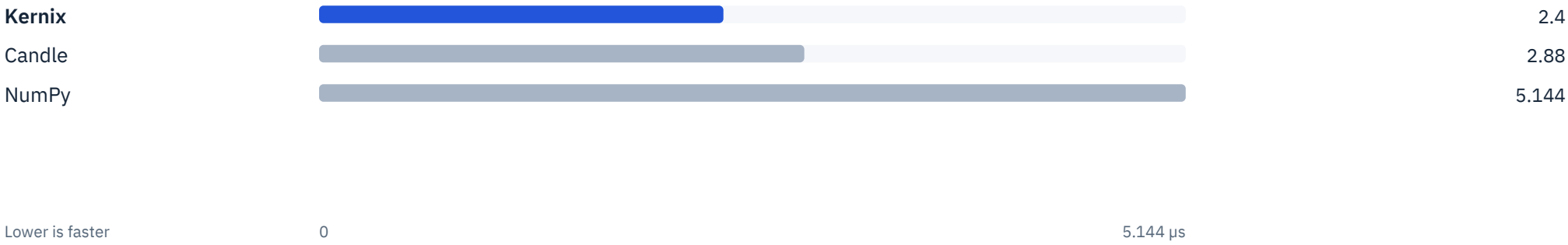
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.11.1 / Explicit parallel execution

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.11.1 / Explicit parallel execution

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



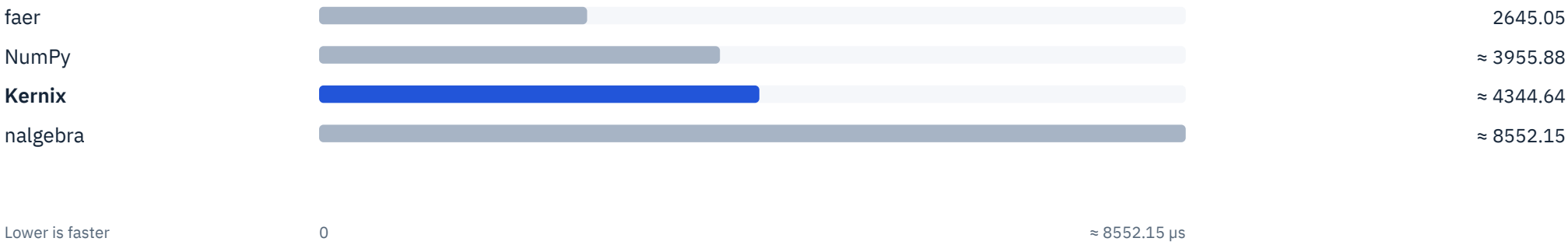
Kernix 0.11.1 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output

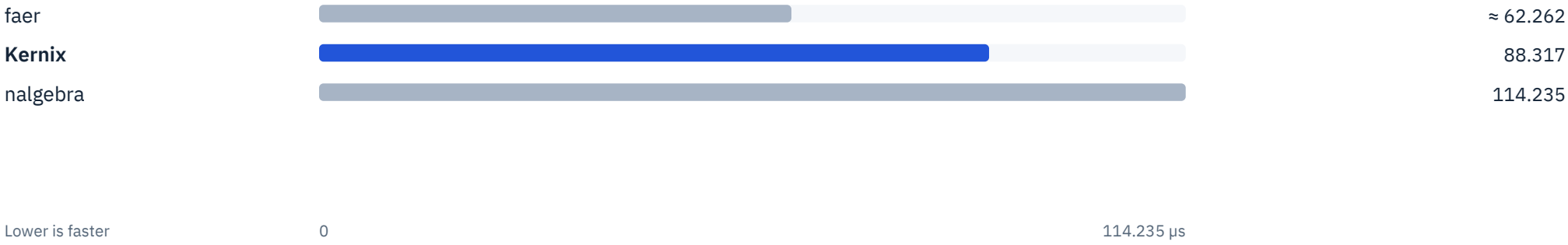
Per-call median



Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.11.1 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

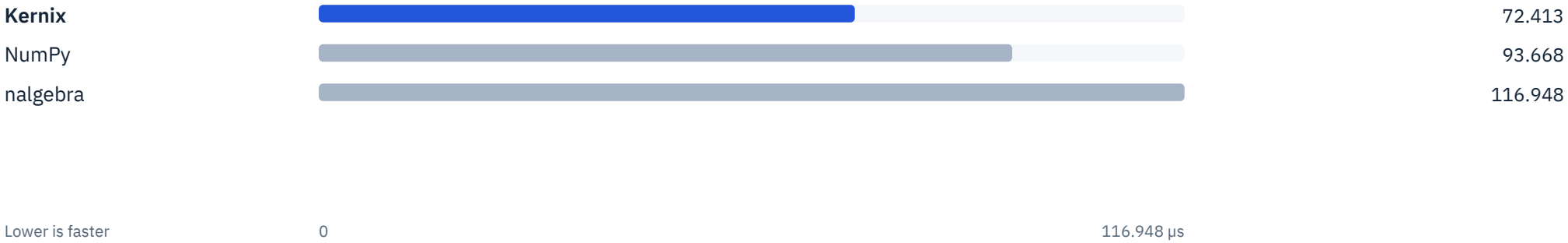
Per-call median



Minimum-norm least squares

64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ

Per-call median



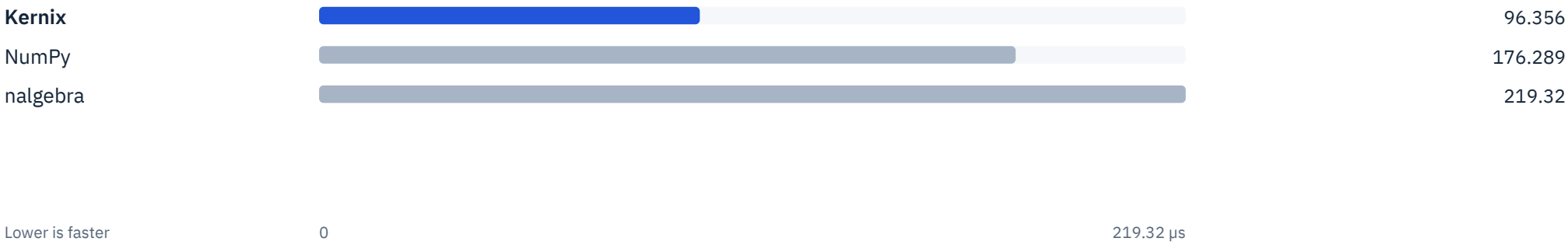
Kernix 0.11.1 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Thin QR

Per-call median

96 × 32 · f64 · 1 worker · allocated output

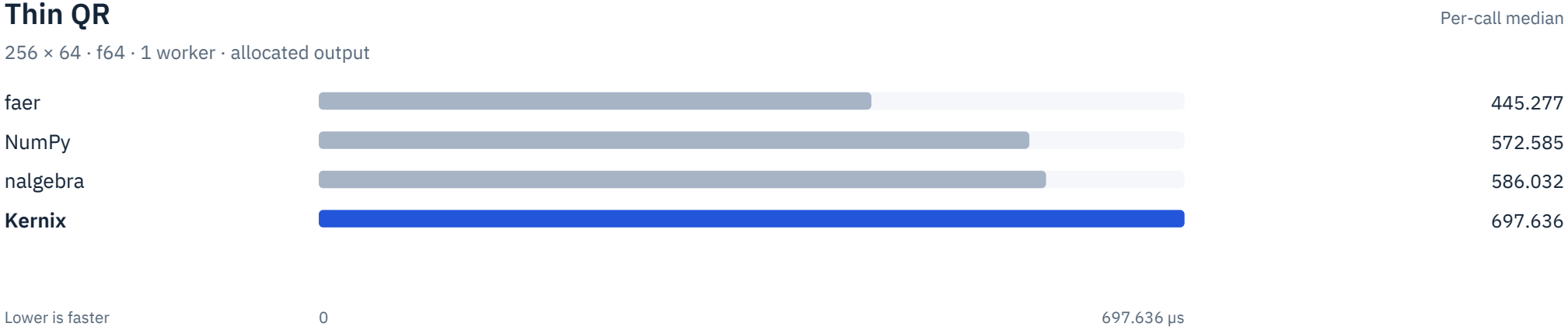


Kernix 0.11.1 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

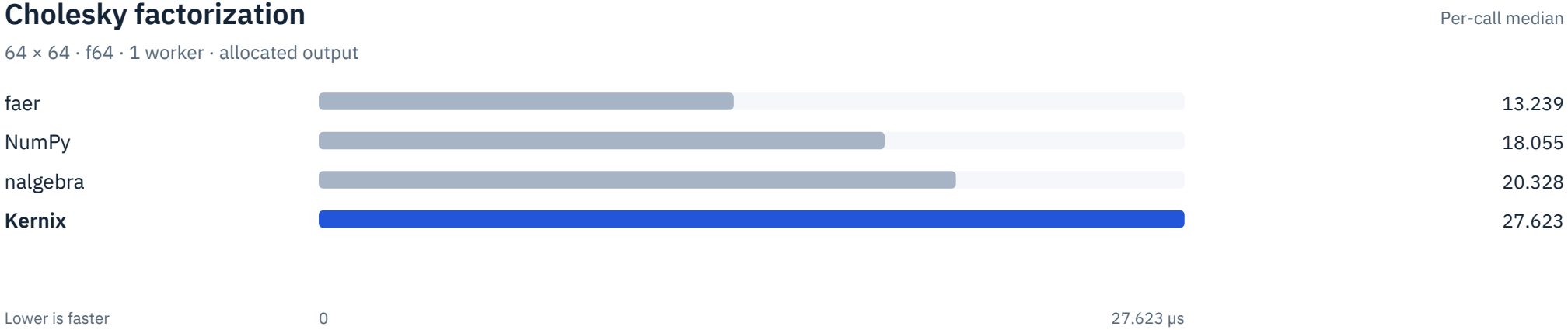
Thin QR

256 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output



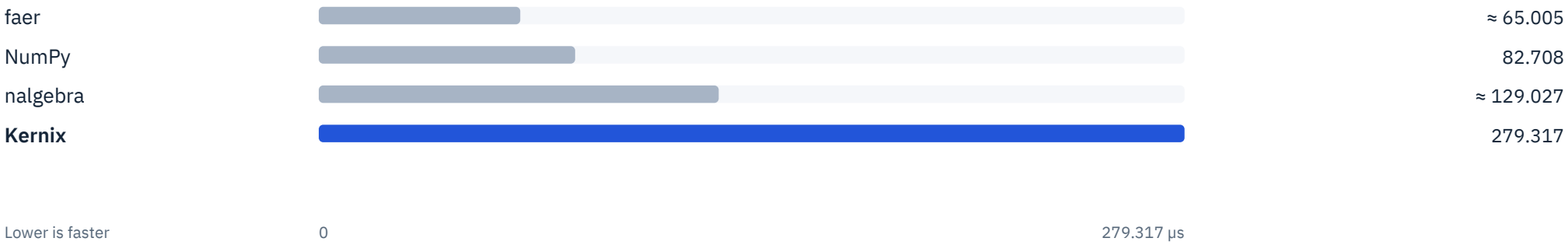
Kernix 0.11.1 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

128 × 128 · f64 · 1 worker · allocated output

Per-call median



Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output

Per-call median



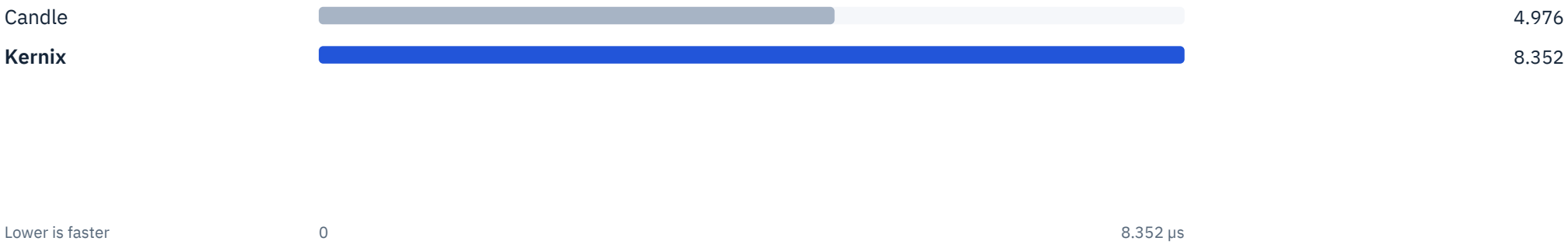
Kernix 0.11.1 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.11.1 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.11.1 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



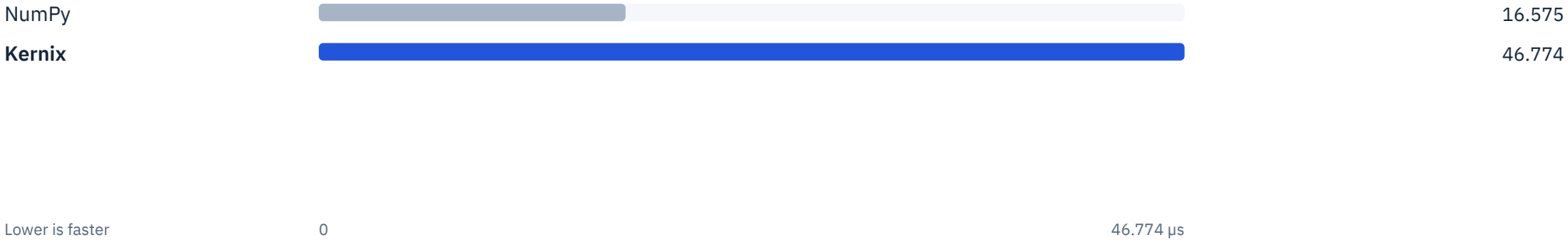
Kernix 0.11.1 / Tensor contractions

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.11.1 / Statistics & activations

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

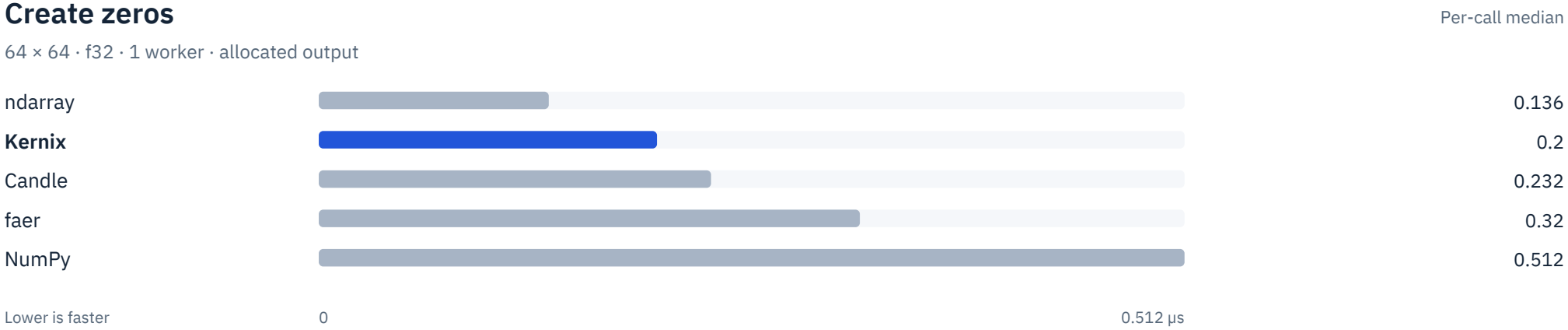


Kernix 0.11.1 / Tensor construction

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output



Create ones

64 × 64 · f32 · 1 worker · allocated output



Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

32 × 32 · 32 × 32 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9975369458128078	0.995085995085995 – 1.002469135802469

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0004237364001505	0.9950299033165496 – 1.0027191413237926

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9988765885739697	0.9974782263525346 – 1.00074901107132

Matrix multiplication

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0009046751669703	0.9990133896044252 – 1.0019451996064153

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0025442864866585	0.997745055627488 – 1.0037790825210626

Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1	0.9977220956719818 – 1.0045766590389016

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9996209159618072	0.9968862813218238 – 1.003373351119016

Independent matrix operands

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0001749760866014	0.9966533344216799 – 1.0017720987711893

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

Timing direction unclear



CURRENT / BASELINE	1	95% INTERVAL	0.9979209979209976 – 1.0020876826722338

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

Timing direction unclear



CURRENT / BASELINE	1.0066889632107023	95% INTERVAL	1 – 1.0067114093959733

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Observed time increase



CURRENT / BASELINE	95% INTERVAL
1.0070253462053007	1.0043716947147794 – 1.010413578148842

Cholesky factorization

256 × 256 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.9936900729098042	0.9904663726385025 – 0.9961985447269444

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.9949989260086665	0.9933922773224124 – 0.9990776670261271

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0075606057275983	0.9775864660039932 – 1.0349407616669952

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9969792279009496	0.9902385219515536 – 1.0019951031910102

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0001844035569398	0.9954224992336773 – 1.0014575763174642

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	195439	195.439
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	169105	169.105
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	100061	100.061
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	85629	85.629
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	3448	3.448
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	3048	3.048
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	3223	3.223
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	3496	3.496
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	8223	8.223
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	3248	3.248

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	21807	21.807
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	31055	31.055
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	22504	22.504
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	22647	22.647
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	22455	22.455
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	22215	22.215
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	194662	194.662
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	208863	208.863
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	168033	168.033
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	168731	168.731

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	209064	209.064
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	168273	168.273
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	1463074	1463.074
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	1321246	1321.246
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	1327137	1327.137
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	1322459	1322.459
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	1536267	1536.267
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	1318796	1318.796
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	11255	11.255
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	20103	20.103

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	12288	12.288
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	11775	11.775
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	11943	11.943
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	11280	11.28
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	93892	93.892
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	86934	86.934
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	85965	85.965
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	94036	94.036
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	85421	85.421
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	99900	99.9

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	756525	756.525
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	722591	722.591
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	766241	766.241
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	673781	673.781
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	661830	661.83
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	751156	751.156
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	4432	4.432
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	12991	12.991
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	3832	3.832
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	2880	2.88

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	5144	5.144
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	2400	2.4
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	16575	16.575
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	92156	92.156
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	698289	698.289
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	48670	48.67
Cholesky factorization / nalgebra 512 × 512 · f64	8552152	8552.152
Cholesky factorization / NumPy 512 × 512 · f64	3955885	3955.885
Cholesky factorization / faer 512 × 512 · f64	2645050	2645.05
Cholesky factorization / Kernix 512 × 512 · f64	4344642	4344.642

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / faer 96 × 32 · f64	62261.99999999999	62.26199999999999
Column-pivoted QR / nalgebra 96 × 32 · f64	114235	114.235
Column-pivoted QR / Kernix 96 × 32 · f64	88317	88.317
Column-pivoted QR / nalgebra 256 × 64 · f64	1149984	1149.984
Column-pivoted QR / faer 256 × 64 · f64	473365	473.365
Column-pivoted QR / Kernix 256 × 64 · f64	784075	784.075
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	116948	116.948
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	93668	93.668
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	72413	72.413
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	176289	176.289

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	96356	96.356
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	219320	219.32
Thin QR / faer 96 × 32 · f64	65165.00000000001	65.16500000000001
Thin QR / NumPy 96 × 32 · f64	66414	66.414
Thin QR / Kernix 96 × 32 · f64	71838	71.838
Thin QR / nalgebra 96 × 32 · f64	61333.00000000001	61.33300000000001
Thin QR / faer 256 × 64 · f64	445277	445.277
Thin QR / nalgebra 256 × 64 · f64	586032	586.032
Thin QR / Kernix 256 × 64 · f64	697636	697.636
Thin QR / NumPy 256 × 64 · f64	572585	572.585

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 64 × 64 · f64	27623	27.623
Cholesky factorization / faer 64 × 64 · f64	13239	13.239
Cholesky factorization / NumPy 64 × 64 · f64	18055	18.055
Cholesky factorization / nalgebra 64 × 64 · f64	20328	20.328
Cholesky factorization / NumPy 128 × 128 · f64	82708	82.708
Cholesky factorization / nalgebra 128 × 128 · f64	129027.00000000001	129.02700000000001
Cholesky factorization / faer 128 × 128 · f64	65005.00000000001	65.00500000000001
Cholesky factorization / Kernix 128 × 128 · f64	279317	279.317
Cholesky factorization / NumPy 256 × 256 · f64	740594	740.594
Cholesky factorization / faer 256 × 256 · f64	403409	403.409

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / nalgebra 256 × 256 · f64	1035177	1035.177
Cholesky factorization / Kernix 256 × 256 · f64	768251	768.251
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	4976	4.976
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	8352	8.352
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	7159	7.159
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3504	3.504
Model inference / Candle batched normalized/sliced MLP · f64	5368	5.368
Model inference / Kernix batched normalized/sliced MLP · f64	3080	3.08
Model recording + forward / Kernix batched normalized/sliced MLP · f64	3856	3.856
Model recording + forward / Candle batched normalized/sliced MLP · f64	7672	7.672

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model backward / Kernix prepared graph → fresh gradients · f64	4184	4.184
Model backward / Candle prepared graph → fresh gradients · f64	21335	21.335
Parameter update / Kernix transactional SGD, fixed gradients · f64	736	0.736
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	46774	46.774
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	16575	16.575
Row softmax / Candle 256 × 64 · f64	110651	110.651
Row softmax / ndarray 256 × 64 · f64	97324	97.324
Row softmax / Kernix 256 × 64 · f64	113124	113.124
Row softmax / NumPy 256 × 64 · f64	133858	133.858
Create zeros / faer 64 × 64 · f32	320	0.32

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / Candle 64 × 64 · f32	232	0.232
Create zeros / NumPy 64 × 64 · f32	512	0.512
Create zeros / Kernix 64 × 64 · f32	200	0.2
Create zeros / ndarray 64 × 64 · f32	136	0.136
Create ones / Candle 64 × 64 · f32	344	0.344
Create ones / ndarray 64 × 64 · f32	208	0.208
Create ones / NumPy 64 × 64 · f32	1528	1.528
Create ones / Kernix 64 × 64 · f32	256	0.256
Create ones / faer 64 × 64 · f32	239.99999999999997	0.23999999999999997